

Date:	DATA VISUALIZATION USING POWER BI
Ex.No:6B	

AIM:

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making using the Earthquake dataset.

PROCEDURE:

1. Start Power BI Desktop application.
2. Load the Earthquake dataset using Get Data -> Excel/CSV.
3. Open Power Query Editor to inspect data types, clean column names, and remove duplicates.
4. Rename and transform columns as needed.
5. Create calculated measures using DAX for Total Earthquakes, Maximum Magnitude, and Average Depth.
6. Add a Bar Chart visual to show magnitude distributions by magnitude type.
7. Add a Pie Chart visual to display alert level contributions.
8. Add KPI Cards for key metrics.
9. Insert a Slicer visual for filtering data by continent/country region.
10. Organize visuals into an interactive dashboard report and save.
11. Stop.

OUTPUT:

Importing Data into Power BI:

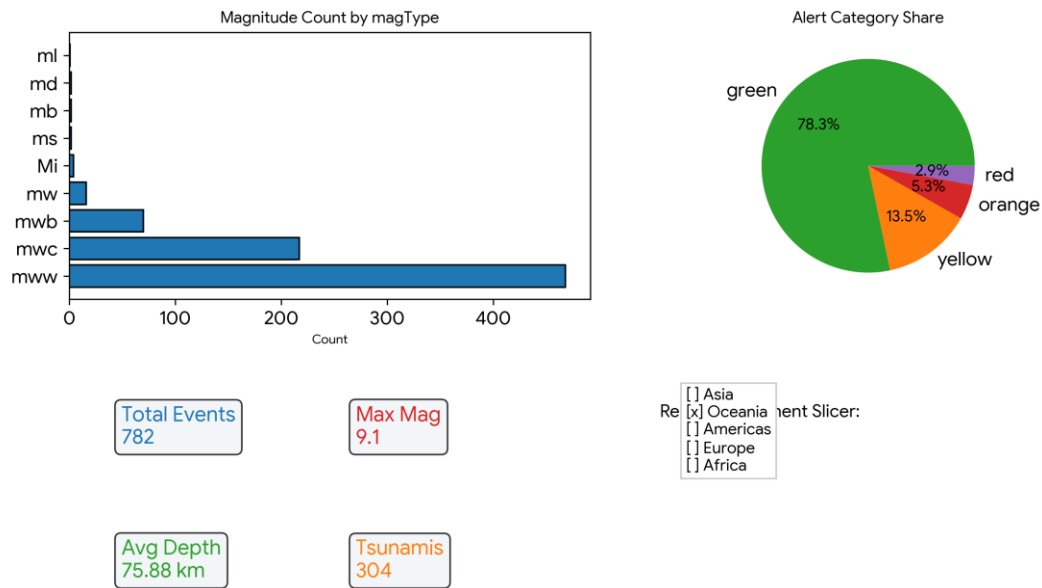
Power Query Editor Loaded: earthquake_data
Columns: title, magnitude, date_time, cdi, mmi, alert, tsunami, sig, net, nst, dmin, gap, magType, depth, latitude, longitude, location, continent, country.

Renamed and Cleaned Dataset Loaded Successfully.

Creating Power BI Visualizations & DAX Calculations:

1. Bar Chart -> Magnitude Type Count
2. Pie Chart -> Alert Category Distribution
3. KPI Cards -> Total Earthquakes (782), Max Magnitude (9.1), Avg Depth (75.88 km), Tsunamis (304)
4. Interactive Slicer -> Filter by Region / Continent
5. DAX Formulas:
 - Total Events = COUNT(earthquake_data[title])
 - Max Magnitude = MAX(earthquake_data[magnitude])
 - Avg Depth = AVERAGE(earthquake_data[depth])

EARTHQUAKE DATA ANALYSIS USING POWER BI



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RESULT:

The Earthquake dataset was successfully imported, transformed, and analyzed using Power BI Desktop. Interactive dashboards and reports were created using different visualizations. These dashboards helped in understanding the dataset clearly and supported better decision-making.